AL-2001
Artificial Intelligence
Lab # 11

Objectives:

- NumPy and Pandas

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. First think about statement problems and then write your logic on Copy / Notebook.
2. Write **Your Name** and **Roll No** on your Paper/Sheet's first page.
3. Do not copy from any source otherwise you will be penalized with negative marks.
4. Complete your lab **within given Time Slot**.
5. Paste all your codes along with screenshots in a word file and renamed with your roll number.
6. Keep all your source files in your computer for verification. Do not overwrite a single source file for all programs.

Problems:

Exploratory Data Analysis and Insights from a Multi-Attribute Dataset

Task Objectives:

1. Data Loading and Initial Exploration:

- **Objective:** Getting to know the dataset.
- **Steps:**
  - Use Pandas to load the dataset (**pd.read_csv()**) and display the first few rows (**df.head()**) to understand the structure.
  - Calculate basic statistics (**df.describe()**) for each attribute: mean, median, standard deviation, min, max.
  - Check data types (**df.dtypes**) to understand how the information is stored.

2. Data Cleaning and Preprocessing:

- **Objective:** Preparing the data for analysis.
- **Steps:**

- Identify any missing values (**df.isnull().sum()**) and decide whether to fill in (**df.fillna()**) or remove these entries.
- If the education level is represented as numerical data, convert it into categories using **df['Education Level'] = df['Education Level'].astype('category')**.

3. Data Exploration and Analysis:

- **Objective:** Understanding relationships between attributes.
- **Steps:**
  - Calculate correlations or associations using **df.corr()** to understand if there's a connection between attributes.
  - Use scatter plots (**plt.scatter()**) or histograms (**plt.hist()**) to visualize how different attributes relate to each other.
  - Group data by age ranges (**pd.cut()**) and analyze how health scores or income levels vary across these groups.

4. Feature Engineering and Analysis:

- **Objective:** Creating new features and understanding their impact.
- **Steps:**
  - Create new features based on existing ones, like dividing ages into groups (**pd.cut()**).
  - Analyze these new features using visualizations or correlations to see how they relate to the original attributes.

5. Statistical Analysis and Insights:

- **Objective:** Delving deeper into relationships using statistical tests.
- **Steps:**
  - Perform statistical tests (e.g., **scipy.stats**) to understand connections between attributes, such as income across different education levels.
  - Derive insights from the tests, like how education levels might impact income or health scores.

6. Visualization and Presentation:

- **Objective:** Illustrating findings for better understanding and presentation.
- **Steps:**

- Create visual representations using Matplotlib or Seaborn for better comprehension and presentation of relationships discovered.
- Summarize key findings and insights in a way suitable for presentations, perhaps using Jupyter Notebooks.

You need to do your exercise within the given time.